HyperionDev

The Programming Life Cycle

Practical Skills Module 1: Topic 5

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Topic Code	PM-01-PS05:
Topic	Apply the programming life cycle to develop a solution

Scope of Practical Skill

Given an applicable instruction and access to a learning platform, the learner must be able to:

- PA0501 Apply the various stages (sequential and reiteratively) in the programming life cycle when developing an appropriate solution

Applied Knowledge

- AK0501 Concept, definition and functions of basic Python functionalities

Internal Assessment Criteria

- IAC0501 Expected results with the programming life cycle in Python are achieved

Resources:

Knowledge	Information from Python Programmer Curriculum
National Curriculum Framework	900221-000-00-PM-01



Practical task

Follow the facilitator's instructions to complete the following activities:

Save the steps followed as a screenshot **LifeCycle**.

Use the following scenario to complete the activities as set out:

Scenario: Developing a Weather Forecasting Mobile App

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A team of developers is tasked with creating a mobile app that provides real-time weather forecasts. To ensure the app is built effectively, they apply the various stages of the programming life cycle, both sequentially and reiteratively, to develop the solution. Here's how each stage is applied:

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1. Problem Definition

Sequential Application:
The team works with stakeholders (e.g., meteorologists and app users) to define the app's purpose and features:

The app should display current weather conditions, hourly forecasts, and a 7-day outlook.

Users should be able to search for locations and set preferences for units (e.g., Celsius or Fahrenheit).

Alerts for severe weather must be included.

Reiterative Application:
Feedback is collected from stakeholders, requiring the team to revisit and refine the problem statement, such as adding accessibility features like voice output for visually impaired users.

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2. Feasibility Study and Planning

Sequential Application:
The team analyses whether the app is feasible within budget, timeline, and resources. They outline:

The technical feasibility of integrating APIs for weather data.

The cost of development tools and hosting services.

A timeline for each development phase.

Reiterative Application:
Unexpected challenges (e.g., the chosen API has limited accuracy) prompt the team to revise their plans and find a better API provider.

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3. System Design

Sequential Application:
The team designs the app’s architecture, including:

Frontend: User interface with intuitive navigation and a visually appealing design.

Backend: API integration for real-time weather data retrieval.

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Database: Storing user preferences and location data.

Reiterative Application:
During testing, users find the interface confusing, requiring the team to redesign the navigation flow for better usability.

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4. Implementation

Sequential Application:
Developers write the code based on the system design.

The backend team integrates weather APIs.

The frontend team develops the user interface.

Reiterative Application:
As bugs are identified during development, developers revisit and rewrite parts of the code to optimize performance and ensure compatibility across devices.

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5. Testing

Sequential Application:
The app undergoes unit testing, integration testing, and system testing to ensure all components work together.

Unit Testing: Checks individual modules like location search functionality.

Integration Testing: Verifies that the weather API delivers accurate data to the frontend.

System Testing: Confirms the app functions as expected on different devices.

Reiterative Application:
Bugs or errors identified during testing are fixed, and the app is tested again. For example, incorrect time zone data requires changes in the API handling code.

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6. Deployment

Sequential Application:
Once testing is complete, the app is released on app stores (Google Play and Apple App Store).

Reiterative Application:
After deployment, user feedback highlights issues, such as slow loading times, requiring developers to revisit the code and release updates.

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7. Maintenance

Sequential Application:
The app is monitored for performance and updated regularly to ensure it remains functional with new devices and operating systems.

Reiterative Application:
New features (e.g., adding radar maps or air quality indicators) are developed and integrated into the app based on user suggestions, restarting the development life cycle.

By applying the programming life cycle both sequentially and reiteratively, the development team ensures the weather app meets user needs, adapts to changing requirements, and evolves over time to remain competitive.

Apply the stages of the programming life cycle to the given Weather Forecasting App scenario. For each of the 7 stages, you must describe:

- How that stage would be applied in sequence (done step-by-step).
- How that stage would be applied iteratively (done again based on feedback, errors, or new requirements).

These stages are:

1. Problem Definition
2. Feasibility Study and Planning
3. System Design
4. Implementation
5. Testing
6. Deployment
7. Maintenance

Complete the activity in a document (e.g., Google Docs or Microsoft Word) and save the document with the filename **LifeCycle** and convert it to a PDF before submission

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Take note

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.

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